Student Record Management System Using Python

1. Introduction

In educational institutions, managing student records manually can be time-consuming and prone to errors. A computerized system helps improve accuracy, efficiency, and data management. This project aims to develop a Student Record Management System using Python.

2. Problem Statement

Manual management of student records can lead to data loss, duplication, and difficulty in retrieving information. There is a need for a simple system that allows users to store and manage student data efficiently.

3. Objectives

- To develop a system to manage student records
- To store student data using file handling
- To allow searching, updating, and deleting records
- To improve data management efficiency

4. Methodology

The system is developed using Python programming language. The program uses functions and file handling to store and retrieve data from a JSON file. A menu-driven interface allows users to perform operations such as adding, viewing, searching, updating, and deleting records.

5. Tools and Technologies Used

Python
JSON File
Visual Studio Code
Windows Operating System

6. Implementation

The program consists of multiple functions:

Add Student
View Students
Search Student
Update Student
Delete Student

Student data is stored in a JSON file named students.json.

7. Results

The system successfully performs all operations related to student record management. It provides a simple and efficient way to store and manage student information.

8. Challenges Faced

Understanding file handling
Managing data storage
Handling user input

9. Conclusion

The Student Record Management System demonstrates how Python can be used to solve real-world problems. It improves efficiency and reduces manual errors in managing student records.

10. Future Scope

Add graphical interface
Use database like MySQL
Add login authentication system